Lab 6: Front-End Frameworks — Building Responsive Portals with Bootstrap 5

Objectives

- **Bootstrap 5 CDN Integration:** Link the Bootstrap 5 stylesheet and JavaScript bundle directly into an HTML document.
- **Responsive 12-Column Grid System:** Use Bootstrap containers, rows, and grid column classes (col-12, col-md-*, col-lg-*) to construct adaptive layouts across mobile, tablet, and desktop viewports.
- **Pre-Built UI Components:** Implement responsive Bootstrap navigation bars (.navbar), hero containers (.p-5, .rounded-3), badges, buttons (.btn), and card containers (.card).
- **Utility-First Styling:** Use Bootstrap utility classes for layout spacing (p-*, m-*), typography (fw-bold, text-muted), background/text colors (bg-*, text-*), shadows (shadow-sm), and borders (border-*).
- **Interactive Components:** Connect trigger elements to interactive Bootstrap components (such as Modals or Collapsible Navbars) using HTML data-bs-* attributes.

Task Overview

Create an original, interactive multi-section webpage for a topic of your choice (e.g., a developer portal, gaming hub, project management dashboard, online course platform, or product showcase). You must construct index.html from scratch using **Bootstrap 5 utility classes and components** without writing custom CSS rules in external files.

Technical Requirements

1. File Setup & Bootstrap 5 CDN Linking

- **Single File Structure:** Create a clean index.html file.
- **Bootstrap CSS CDN:** Link the Bootstrap 5 stylesheet inside the <head> tag.
- **Bootstrap JS Bundle CDN:** Link the Bootstrap 5 JS bundle script right before the closing </body> tag to enable interactive elements.
- **Zero Custom CSS / Inline Styles:** All styling, spacing, layout structure, and color formatting must be accomplished exclusively using Bootstrap utility classes and components.

2. Responsive Navigation Header (.navbar)

- Implement a <nav> bar using Bootstrap's .navbar component family.
- Configure a sticky or top-fixed navigation bar featuring a brand/logo title and navigation links.
- **Mobile Collapse Feature:** Use the .navbar-toggler button along with data-bs-toggle="collapse" and data-bs-target so navigation links automatically collapse into a hamburger menu on mobile screens.

3. Responsive 12-Column Content Grid (.container, .row, .col-*)

- Wrap the main content inside a Bootstrap .container with vertical spacing utilities (e.g., my-5 or py-4).
- Create a grid section using a .row container housing at least **3 content cards** (.card elements).
- **Multi-Breakpoint Column Rules:** Every card column must include breakpoint classes to dynamically adjust column spans across device sizes:
    - **Mobile (<768px):** Full-width stack (col-12 — 1 card per row).
    - **Tablet (≥768px):** Half-width grid (col-md-6 — 2 cards per row).
    - **Desktop (≥992px):** One-third width grid (col-lg-4 — 3 cards per row).

4. Card Styling & Utility Classes

- Structure card content using .card, .card-body, .card-title, and .card-text.
- Use utility classes for styling across all cards:
    - **Equal Height Alignment:** Apply .h-100 to force cards within the same row to match height.
    - **Spacing Utilities:** Use margin and padding classes (p-3, mb-2, my-3).
    - **Visual Accents:** Apply drop shadows (shadow-sm or shadow), rounded corners (rounded-3), badges (.badge), and contextual border/color utilities (border-top, border-primary, bg-light, text-muted).

5. Interactive Component (Modal or Dropdown)

- Include at least one interactive Bootstrap JavaScript component (e.g., a Modal popup window or Dropdown menu).

- Connect a call-to-action button to the interactive target using data-bs-toggle and data-bs-target attributes without writing any custom JavaScript code.

Part 2: Assessment & Grading

Lab 6 Grading Rubric (100 Points Total)

Evaluation Criteria	Point Value	Key Checklist Items
CDN Integration & Setup	**20 pts**	Valid Bootstrap 5 CSS and JS Bundle CDN scripts included. HTML structure built cleanly without custom <style> blocks, inline style="" attributes, or external CSS stylesheets.
Responsive Header (.navbar)	**20 pts**	Correct usage of .navbar, .navbar-brand, .navbar-nav, and a functional mobile hamburger menu toggle using data-bs-toggle="collapse".
Responsive 12-Column Grid	**20 pts**	Main layout structured using .container, .row, and multi-breakpoint column classes (col-12, col-md-6, col-lg-4) that adapt smoothly across viewports.
Components & Utility Styling	**20 pts**	Proper application of .card components (.card-body, .h-100), spacing utilities (m-*/p-*), typography (fw-bold, text-muted), and color/border utilities (bg-*, border-*).
Interactive JS Component	**20 pts**	Functional implementation of a Bootstrap interactive component (e.g., Modal or Dropdown) triggered via data-bs-toggle and data-bs-target data attributes.
Total	**100 pts**	

Submission Checklist

- [ ] Project consists of a single index.html file referencing Bootstrap 5 via CDN.
- [ ] No external .css files, <style> tags, or inline style="" attributes are present.
- [ ] Page includes a responsive Bootstrap navigation bar that collapses into a hamburger toggle on mobile screens.
- [ ] Main content utilizes a Bootstrap 12-column grid (container →row →col-*) configured for 1-column mobile, 2-column tablet, and 3-column desktop viewports.

- [ ] Cards use .h-100 alongside Bootstrap spacing, shadow, border, and badge utility classes.
- [ ] Clicking a button triggers an interactive Bootstrap component (such as a Modal popup) using data-bs-* attributes.